

SAIM HASHMI

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EDUCATION

Carleton University

Ottawa, ON

Bachelor of Engineering (B.Eng.), Computer Systems Engineering, Fourth-Year Standing

Expected May 2027

Relevant Coursework: Operating Systems, Computer Networks, Signals & Communications, Data Structures & Algorithms, Object-Oriented Programming, Embedded Systems

TECHNICAL SKILLS

Languages: C, C++, Go, Java, JavaScript, Python, Ruby, SQL, TypeScript

Frameworks: FastAPI, Flask, Next.js, Node.js, React.js, ROS2, Ruby on Rails, Scikit-learn, Tailwind CSS

Developer Tools: AWS, Bash, Docker, GCP, Git, GitHub, GitLab, Grafana, Jenkins, Jira, Linux, pytest, QNX, Tmux, Vim

AI Tools: Aider, Claude Code, CodeRabbit, Cursor, GitHub Copilot, Omniparse

RELEVANT EXPERIENCE

Synopsys | Azure DevOps, Bash, CI/CD, Docker, Python, Cypress, Plotly, React

Remote

Software Development Intern

May 2026 – August 2026

- Built a new plotting feature for Ansys Fluent's visualization tool in **2 weeks**, extending its backend API and shipping it end-to-end as part of a major customer release used by all Fluent users
- Extended the feature into three of Ansys's Python client libraries (PyFluent-Visualization, PyEnSight, PySimAI) to prevent silent failures when accessed via the Python scripting interface, adding new CI and Cypress test coverage to catch regressions
- Led a Python 3.13 baseline consolidation across **13+** repositories, resolving failing CI builds with DevOps, Fluids, and Fluent Prime

Shopify CLI (OSS) | TypeScript, Ruby, oclif, Node.js, GitHub Actions

Remote

Open-Source Contributor

November 2025 – Present

- Built a new CLI command (`config:pull`) that auto-detects existing project config, skipping repeated setup prompts for users
- Standardized config validation across core commands to prevent inconsistent state and avoid common config-related failures
- Refactored shared configuration modules to unify core logic, reducing duplication and long-term maintenance overhead

CuHacking | Payload CMS, React, TypeScript, Node.js, REST APIs

Ottawa, ON

Full Stack Developer: Website, Dev-Docs, Portal

September 2024 – April 2026

- Engineered a developer portal for **400+ attendees** using modular React, ensuring reliability during live events at scale
- Led the adoption of Storybook for reusable UI components, eliminating redundant testing and per-platform variants site-wide
- Established an Nx monorepo unifying the docs, website, portal, and Figma scopes, clarifying ownership and build hierarchy

Autonomous Robotics Carleton | ROS2, Python, C++, Linux

Ottawa, ON

Software Development Lead: Systems & Tooling

July 2025 – January 2026

- Led ROS2 perception and navigation development in C++/Python, applying concurrent programming to cut inter-process latency
- Established Docker-based CI, enforced code reviews, and expanded test coverage across contributor PRs as codebase scaled
- Defined contribution standards and maintained centralized docs to onboard and support **20+** new contributors effectively

PERSONAL PROJECTS

BOOPBOT | QNX SDP 8.0, ROS2, Qt Quick/QML, C++, Raspberry Pi 5

May 2026 – Present

- Built the Qt Quick/QML terminal-themed UI for a photo booth robot, with a live camera feed and API controls for live event use
- Integrated a Raspberry Pi camera via the QNX Sensor Framework, then resolved a Qt6/ROS2 CMake conflict with a concurrent QLocalSocket bridge and crash-recovery watchdog
- Ported a SLAM stack to QNX Neutrino, resolving thread scheduling conflicts between the SLAM pipeline and QNX's hard deadlines
- Built motor control with encoder feedback and logging, validated using a Logitech controller before integrating the autonomy stack
- Secured a **\$1,000** corporate development grant from BlackBerry QNX to build the robot for cuHacking 7

SkyTalk (SKYTRAC Hackathon) | React Native, Fastify, WebSocket, Prisma, PostgreSQL, DigitalOcean

October 2025

- Built a cross-platform Android/iOS walkie-talkie app (Expo) with AES-256-GCM encrypted push-to-talk over Iridium Certus satellite (**22 kbps, 500–1500ms latency**)
- Built the client-side comms library (ForbiddenLANComms) handling the full push-to-talk connection over an unreliable satellite link
- Implemented timestamp-based floor-control arbitration to resolve simultaneous push-to-talk conflicts before server confirmation, keeping audio responsive at satellite latency

VaultPi | AWS, Terraform, Ansible, Docker, Immich, Tailscale

September 2025 – Present

- Built a personal cloud NAS on a Raspberry Pi, provisioned with Terraform and Ansible across **11** roles for reproducible deployment
- Built a cloud-native, event-driven AI pipeline where S3 uploads trigger a Lambda function calling AWS Bedrock for photo tagging, embedding, and duplicate detection, with results stored in DynamoDB
- Deployed Immich as a self-hosted photo app, replacing iCloud for **14,000+** photos (**300+ GB**) with no storage cap or subscription
- Configured Tailscale for secure remote access to the NAS from anywhere, with no open ports or VPN to maintain